Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka

INDIVIDUAL THESIS

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Individual component: [AI-Driven Cultivation Recommendation System for Small-Scale Herbal Farmers]

Group ID: [R26-IT-021]

Report submitted in partial fulfilment of the requirements for the

[BSc (Hons) in Information Technology]

Department of [Department name]

Sri Lanka Institute of Information Technology

Sri Lanka

[Month Year]

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ABSTRACT

Medicinal and Ayurvedic herbal plants cultivation is one of the most valuable prospects for small farmers in Sri Lanka. However, the choice of a proper herbal plant to cultivate requires careful consideration of numerous factors such as the type of soil, availability of land, climate in the area, budget allocated for cultivation, demand on the market, economic viability, and sustainability, among others. The currently available methods usually take into account particular agricultural aspects without providing a holistic solution to choosing a plant for cultivation according to farmer's conditions. This study attempts to solve the problem with the help of an AI-based cultivation recommendation system. In the suggested system, machine learning techniques are used to determine the correlation between the growing conditions and the medicinal plants suitable for these conditions. The system takes real user input like soil type, land size, and growing budget, and it also considers weather information based on the user's location. The recommendation capability is developed using Flutter-based user interface along with the Python Flask backend. The machine learning models are compared using accuracy, precision, recall, and F1 score to select the most appropriate prediction model. This module is developed as a component of the overall intelligent Ayurvedic medicinal herb plants identification and cost analysis system and acts as the decision support tool for the small-scale farmers. The individual contributions include data set collection, feature extraction, modeling, evaluation, backend implementation, and user interface development.

Keywords: [Medicinal plants, Cultivation recommendation, machine learning, small-scale farmers]

ACKNOWLEDGEMENTS

I would like to express my sincere appreciation to my guides, Mr. Nelum Amarasena and Ms. Kaushika Kahatapitiya, for their guidance, helpful comments, support, and academic assistance throughout this research component. Their knowledge and suggestions played an important role in completing this study.

I would like to express my gratitude towards the academic personnel of SLIIT (Sri Lanka Institute of Information Technology) for providing me with the knowledge and resources necessary for this study.

My special thanks go to the research team for their cooperation and useful suggestions to the project titled “Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka.”

I would like to acknowledge those who offered their suggestions and support throughout the requirement analysis, implementation, testing, and evaluation of the solution.

I also owe my gratitude to my family members and friends who have always encouraged me throughout my academic career and this research project.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
F1	F1-score
ML	Machine Learning
RF	Random Forest
UI	User Interface
UX	User Experience

Table 1: List of abbreviations

1 INTRODUCTION

1.1 Overall project context

The name of the complete research project is “Intelligent Ayurvedic Herbal Plant Identification and Economic Analysis in Sri Lanka.” The main objective of the project is to use intelligent and machine learning technologies for the identification, growing, and economic analysis of Ayurvedic and medicinal plants in Sri Lanka.

It has been seen that Sri Lanka has a rich tradition in utilizing medicinal plants through Ayurveda. However, small-scale farmers could face many difficulties in choosing the appropriate medicinal plants for their cultivation due to certain factors, including soil condition, climate, availability of land, cultivation cost, market demand, and other related aspects.

Overall, the objective of the research project is to use intelligent technology in order to assist users in managing medicinal plants.

1.2 Individual component context

The individual component that is developed through this research is AI-Driven Cultivation Recommendation System for Small-Scale Herbal Farmers.

The role of this component is to provide recommendation about suitable herbal or medicinal plants depending on conditions and constraints of the particular farmer rather than just suggesting a general list of herbs. The system takes into account various factors concerning both the farmer and the environment of cultivation.

Inputs provided by the user are the following:

Soil type

Available land size

Budget for cultivation

The system will also be able to consider environmental factors such as weather conditions as well as plant-related factors such as cultivation requirements, market demands, economic viability, and sustainability. The component works using machine learning algorithms.

1.3 Problem statement

Some critical decisions have to be made by the small-scale herbal farmers while choosing the medicinal plants for cultivation. The plant that is ideal in one area may not necessarily be ideal in another area due to differences in environmental conditions, availability of land, finances, and marketing opportunities.

The traditional decision-making approach may rely largely on personal experiences or general agricultural knowledge. While this approach is important, it is difficult to take into consideration all these factors at once.

For instance, the farmer may have the appropriate land and soil conditions but lack sufficient money to cultivate a certain plant. Likewise, the plant can be cultivated under certain environmental conditions but lack market demand in the area of focus.

Thus, there is a need to develop an intelligent system that will integrate all these factors and offer personalized suggestions for cultivation of small-scale herbal farmers.

1.4 Research gap

Traditional systems for recommending agriculture usually take into account soil, weather, requirements of crops, and nutrients. Still, many approaches only take into consideration conventional crops and do not focus on medicinal and Ayurvedic herbs that could be grown in the environment of Sri Lanka.

Additionally, there are other requirements from farmers who operate in small scales besides environmental feasibility. These requirements include availability of land, budget, and demand and profitability of the process.

The second limitation is associated with manual collection of technical information about environmental parameters. Instead, the proposed component would help to provide farmers with a more pragmatic approach to enter understandable information, such as soil type, size of the area, and budget, while receiving environmental data using location-based weather information.

Thus, the gap is in existence of the lack of an integrated AI/ML based approach to cultivation recommendations for herbal small-scale farmers.

1.5 Aim and objectives

Aim

The primary aim of this study is to design and test the performance of an AI-based cultivation recommendation system which will help small herbal farmers select the right kinds of medicinal plants taking into account various factors like environmental, land, financial, market and sustainability considerations.

Objectives

- To determine the key factors that play role in the suitability of medicinal-plant cultivation.
- To create a well-defined data set which includes important details about cultivation, environment and economy.
- To build a machine learning model which can recommend the right herbal plants.
- To use location-specific weather details in the recommendation process of cultivation.
- To design and build an interface that allows input of soil type, land size and cultivation budget.
- To test the machine learning model using accuracy, precision, recall and F1-scores.
- To combine the recommendation module with other modules of the whole research project.
- To validate the usability of the recommendation system developed.

1.6 Research questions or hypotheses

These are the research questions guiding the research:

RQ1: What are the most crucial aspects to consider while recommending appropriate herbal plants for small-scale farmers?

RQ2: How well can machine learning approaches help in recommending appropriate herbal plants based on cultivation and environmental factors and other farmer attributes?

RQ3: In what way can weather data from locations make herb cultivation more feasible?

RQ4: How successfully can an AI-based recommendation system be incorporated into an entire Ayurvedic herbal plant information system?

1.7 Scope limitations and assumptions

The scope of this component is restricted to giving cultivation recommendations for medicinal and herbal plants.

The system considers the personal inputs of the farmers such as type of soil, land area, and budget for cultivation. Information about the environment, especially weather conditions, is also taken into account in the recommendation system.

It should be noted that the system does not try to replace agricultural experts or ensure success in cultivation. This will depend on the nature, completeness, and accuracy of the data set available.

The component does not make use of IoT-based soil sensors. Manual sensor-based inputs like soil moisture and soil pH are not treated as mandatory inputs of the user.

Currently, the system is also constrained by the availability of appropriate datasets regarding medicinal plants in Sri Lanka. Some of the advanced functionalities such as automatic soil image processing or sunlight estimation require additional trained models and datasets.

1.8 Individual contribution

This involves the definition and analysis of the problem of cultivating recommendations, determination of suitable features, creation of cultivation data set, development of machine learning models, evaluation of models, implementation of the backend, Flutter frontend development, input validation, developing recommendation API, and incorporation into the entire research system.

This particular component is also responsible for conducting research on how to incorporate farmer constraints with environmental and economic information to help in cultivation decisions.

1.9 Thesis structure

Chapter 1 contains an introduction to the research project, component, problem, research gap, research aims, research questions, scope of research, and individual contributions to the research.

Chapter 2 contains a literature review and related work in the fields of agricultural recommendation systems, machine learning, medicinal plants, and cultivation decision support.

Chapter 3 contains information on the research methodology, data set, experiment method, performance metrics, data analysis, ethical considerations, and reproducibility of the research.

Chapter 4 includes requirements, system architecture, alternative designs, and detailed design of the solution.

Chapter 5 describes the solution implementation, integration, tests performed, development environment, and deployment process.

Chapter 6 includes research results, technical results, validation of results, and integration results.

Chapter 7 interprets the results and draws comparisons with related research.

Chapter 8 describes commercialization, sustainability, scalability, and the Sustainable Development Goals.

Chapter 9 summarizes the research and future development of the research.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature review will examine literature related to machine learning agricultural recommendation systems, crop suitability prediction, medicinal plant cultivation, agriculture decision support systems, weather-based cultivation, economic feasibility, and intelligent recommendation systems.

This literature review will examine literature from research articles, journals, agricultural sources, and machine learning software documentation.

Key topics that will be investigated include:

1. Machine learning in agriculture recommendation systems.
2. Prediction of crops and plant suitability.
3. Medicinal and herbal plant cultivation.
4. Decision support in agriculture based on weather.
5. Market based agriculture recommendation systems.
6. Agriculture decision support systems for farmers.

The final research report must provide details about the database used and the sources examined in this literature review.

2.2 Domain and theoretical foundations

Artificial Intelligence

Artificial Intelligence is a term that describes computational techniques that allow systems to undertake activities that typically need human intelligence such as prediction, classification, decision-making, and pattern recognition.

In this study, artificial intelligence is employed to assist in making decisions about planting by considering various features and recommending plants.

Machine Learning

Machine Learning is a field within artificial intelligence that allows computational models to learn patterns from data and make predictions using previous examples.

Cultivation recommendation can be formulated as a supervised learning problem that uses cultivation related input features to recommend appropriate plants.

Classification

Classification refers to a machine learning problem where the input is classified into any of a few predefined categories.

In our proposal, the types of plants can be used to represent the recommendation classes. The model will learn how to establish associations between the input variables and the appropriate plant class.

Random Forest

Random Forest is a type of ensemble machine learning technique that involves the combination of several decision trees to generate predictions. This is a powerful method for making classifications using structured data having different feature types.

Random Forest is one of the machine-learning techniques that we will consider for the cultivation recommendation.

Agricultural decision support

An agricultural decision support system helps the farmers make better decisions by offering them useful information or suggestions.

The suggested decision support system uses the idea in the field of medicinal plants farming.

2.3 Related research

Several studies have shown the application of machine learning in recommending suitable crops by considering the environment and its various attributes like soil, climate, rain, temperature, and many others.

In some of the agriculture recommender systems, crop suitability based on environmental factors is the main objective. Research has been done on the prediction of diseases, yield, crop classification, and agricultural decision-making.

Nevertheless, the component under discussion is different from all this because it concentrates on medicinal and herbaceous plants and considers the constraints of individual farmers like availability of land and budget.

This study also recognizes the significance of economic factors. Just because an agricultural product is suitable for an environment does not mean that it will be economical for a particular farmer.

So, an attempt has been made in this paper to shift the concept of environmental crop recommendation towards cultivation decision-making.

2.4 Existing systems products and techniques

Current agriculture systems may be classified into different groups.

The first type includes rule-based systems. These systems use pre-defined rules for agriculture to suggest crops. They may be simple to comprehend yet hard to manage when many variables are included.

The second type includes machine learning recommendation systems. These systems learn relationships from historical data and allow for automated prediction.

The third type includes weather-based agricultural apps. They include information on temperature, precipitation, humidity, etc., yet fail to give personalized suggestions for medicinal plants.

The fourth type is represented by agricultural market and economic tools. They can give financial information yet fail to incorporate environmental viability with farmers' limitations.

The suggested component includes all these ideas into one recommendation system.

2.5 Gap synthesis and conceptual framework

The conceptual framework to be designed will comprise four main layers.

Input layer: The farmer will provide the soil type, land size, and budget he/she has.

Environmental layer: Environmental details including location-based weather details like temperature, humidity, winds, sunlight and so forth may be included.

Knowledge/data layer: Information including crop cultivation details, economical information, market demands, and sustainability information will be provided by the system.

Recommendation layer: Based on the information, a machine learning algorithm will recommend crops.

Thus, the conceptual framework combines farmer information, environmental information, and crop information using AI algorithms.

3 RESEARCH METHODOLOGY

3.1 Research design

This research employs an applied research and software development approach.

The research process entails:

1. Requirement identification
2. Literature review
3. Dataset preparation
4. Feature identification
5. Data preprocessing
6. Machine-learning model development
7. Model evaluation
8. Backend development
9. Flutter interface development
10. System integration
11. Functional testing
12. Validation and analysis

This research is a blend of machine learning evaluation and software development.

3.2 Data participants or study materials

The main source for the study will be the medicinal-plant cultivation dataset for training and testing the recommendation model.

The dataset includes information about herbal and medicinal plants and their cultivation features. Such features will be environmental suitability, requirements for cultivation, and economic/market-oriented features.

The dataset size, number of features, number of plant classes, and the source of the data must be indicated depending on the final dataset that was used in the implementation.

If there is feedback from the stakeholders/farmers, information about the participants must also be recorded.

3.3 Experimental or study procedure

This experimental procedure has a number of steps.

First, the factors influencing cultivation suitability are determined via a literature survey and domain analysis.

Secondly, the data related to cultivation is prepared and preprocessed.

Thirdly, the relevant features are chosen and converted to be input for machine learning algorithms.

Fourthly, machine learning algorithms are trained with the help of these features.

Fifthly, machine learning models are assessed using accuracy, precision, recall, and F1-score.

Sixthly, the optimal model is chosen based on the results of the assessment and real needs.

Lastly, this model is incorporated into the Flask backend, and the Flutter application interacts with this backend and outputs the recommendation results.

3.4 Measures and evaluation metrics

The metrics include accuracy, precision, recall, and F1-score.

Accuracy

Accuracy is the percentage of total correct predictions made by the model.

Accuracy = Correct Predictions / Total Predictions

Precision

Precision is the ratio of the total predicted instances that belong to a class to the total number of predicted instances for the class.

Precision = True Positives / (True Positives + False Positives)

Recall

Recall is the effectiveness of the model in determining true positive values.

Recall = True Positives / (True Positives + False Negatives)

F1-score

F1-Score is the combination of precision and recall.

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

These metrics give a better idea of model performance beyond just accuracy.

3.5 Data analysis

Data Analysis includes examining the dataset, discovering any missing data, ensuring data consistency, variable transformation, data encoding when necessary, and finally getting the dataset ready for machine learning algorithms.

The dataset is split into trains and test parts to check the performance of the model learned on the data that has never been seen before.

Performance of the candidate models is compared based on the evaluation metrics chosen.

The real outcome of the experiments should be reported in the final thesis.

3.6 Ethics privacy security and responsible AI

The system is intended to gather as little personal data as possible.

The recommendation system must not make any promises that are false and ensure the user profits financially or will succeed in cultivation.

The system should also account for the potential issue of dataset bias due to either the lack of data or its geographical location.

Any external datasets, APIs, images, research papers, and documentation utilized in the development of the project should be referenced properly.

API keys and backend credentials should not be exposed in the source code.

3.7 Reproducibility and research artefacts

Research artifacts consist of the cultivation data set, source code, machine learning model, data pre-processing steps, API implementation, Flutter application, test cases, and evaluation results.

Backend has been written in Python using Flask library and Frontend has been written in Flutter.

A final research report should have enough details about the development environment and dependencies required to reproduce the implementation.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Functional requirements

The system will:

1. Enable the user to specify the soil type
2. Enable the user to specify the available land size
3. Enable the user to specify the available cultivation budget
4. Check the validity of user input data
5. Retrieve weather data according to location when available
6. Process the cultivation-related inputs
7. Make appropriate recommendations regarding herb plants
8. Show recommendations to the user
9. Facilitate communication between the Flutter frontend and Flask backend
10. Fit into the bigger research solution

Non-functional requirements

The system must be:

1. Easy to use
2. Provide recommendations that can be understood
3. Give responses within a reasonable time
4. Check for invalid inputs
5. Maintainable
6. Provide reliable API communication
7. Keep secure the configuration data
8. Extendable in terms of the plant database

4.2 Component architecture within the complete solution

Proposed component is in line with the client-server architecture.

Flutter front-end provides the user interface.

Flask back-end takes the request from the front-end and does the recommendation processing.

The **machine-learning** model provides the predictive recommendation functionality.

Cultivation dataset provides the plant and cultivation data necessary for the model.

Weather data service can provide environmental data based on the location.

Recommendation component works together with other components in the research project.

4.3 Design alternatives and decisions

A rule-based approach was chosen due to the direct encoding of agricultural rules. Nevertheless, a pure rule-based system may become too complicated when more than one variable is involved.

A machine learning model was chosen since it allows one to learn patterns based on the structured data, and one can evaluate its performance quantitatively.

Flutter was chosen as the front-end since it allows cross-platform application development.

Flask was chosen as the backend since it offers a light-weight framework that allows the development of machine-learning functionality via API.

The system does not require IoT sensors since the topic of the research is AI/ML-based decision making.

4.4 Detailed design

The first step in recommending is when the user launches the cultivation recommendation interface.

The user provides the soil type, available plot size, and cultivation budget.

Validation takes place for the values provided.

The budget is expressed in numeric Sri Lankan Rupees and negative numbers are not accepted.

The frontend sends the information to the backend API.

The information is processed on the backend and prepared for the recommendation model.

The recommendation is generated using machine learning.

Finally, the recommendation is sent back to the Flutter app.

4.5 Standards and professional practice

Implementation adheres to software engineering guidelines like modularity, segregation of duties, input validation, API communication, maintainability, and testing.

The assessment of machine learning is based on classification measures.

References to outside sources must adhere to the academic referencing style.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The front end of the system will be programmed using the Flutter framework.

The back end will be programmed using Python Flask framework.

The machine learning capabilities will be done through machine learning libraries using Python.

The cultivation dataset will be kept in CSV file.

The back end will be created using the Python virtual environment.

The project structure will include the back end and cultivation dataset.

5.2 Component implementation

The recommendation interface for cultivation has been developed in the Flutter framework.

The interface includes the following fields for entering the type of soil, available land size, and how much money the farmer wants to invest in cultivation.

The investment field has been developed as numeric entry and uses Sri Lankan Rupee formatting if necessary.

Budget cannot have negative values.

There is an API request from frontend to Flask backend.

Backend processes the entered information and forwards the corresponding values to the recommendation engine.

The result of the recommendation is sent back to the frontend.

5.3 Integration with the other components

Recommendation system is only one part of the whole research project.

The entire system concentrates on intelligent plant recognition and economic analysis in the context of Ayurvedic herbs.

The part that concerns cultivation recommendations works as an additional function that assists the user in choosing appropriate plants to cultivate.

Backend API ensures clear communication between the recommendation model and Flutter app.

5.4 Verification and software testing

Testing is carried out at various levels.

Input Validation Testing

Testing is conducted on valid and invalid values for soil type, land size, and budget.

Boundary testing

Boundary values for budget and land size are tested to make sure the system can handle minimum and maximum values.

API testing

Testing is conducted on the backend API to make sure that the system responds to valid requests appropriately.

Model testing

The machine learning model is tested with the chosen metrics.

Integration testing

Integration testing of the whole process from receiving data to displaying results is performed.

Error handling

Invalid input, empty input, error response from API, and lack of information should not crash the application.

5.5 Deployment and operation

The Flask backend is run locally during development.

The Flask backend is launched via the project's Python virtual environment as well as Flask app entry point.

The Flutter front-end connects to the backend at the specified URL.

In production, the backend can be deployed using a cloud service, and the front-end can be distributed as a mobile/web application.

6 RESULTS AND VALIDATION

6.1 Research results

Performance metrics to evaluate the proposed recommendation model include accuracy, precision, recall, and F1-Score.

Final experimental outcomes are to be placed on the following table.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.00%	0.00%	0.00%	0.00%
Model 2	10.00%	7.50%	10.00%	8.13%
Model 3	6.00%	4.67%	6.66%	5.00%

Table 2: Final experimental outcomes

It is recommended that the model having the best performance be chosen based on the findings of the experiment.

It should also be taken into account whether the model has consistent performance for all types of plants.

6.2 Product and technical results

The developed system successfully delivers a user interface where users can enter information related to the cultivation process as well as communicate with the Flask backend.

The system features:

- Soil-type selection
- Land-size input
- Budget input
- Input validation
- Communication with API
- Prediction through machine learning
- Recommendation display

- This section requires screenshots of the user interface

6.3 User or stakeholder validation

The validation by the user/stakeholder is needed to establish that the system is understandable and useful to the target audience.

This includes:

- Ease of data input
- Clarity of recommendation
- Perceived usefulness
- Understandability
- Satisfaction

The findings in the final report will indicate the real number of subjects as well as the results obtained.

If no user validation was done formally till now, then there is no need for fictitious numbers in this section.

6.4 Integration results

The integration test examines the communication between the Flutter frontend and the Flask backend.

The full workflow is:

User inputs → Validation → API request → Backend Processing → Prediction via ML → Response from API → Output recommendations

The test proves successful integration when the valid input data travels the entire way and produces the desired recommendation output.

6.5 Summary of findings

Feasibility of using machine learning for the decision-making process for herbal-plant farming is shown by the developed component.

This system integrates both practical input from farmers and machine learning suggestions and is supposed to function within the overall intelligent information system for Ayurvedic plants.

It is necessary to base the final quantitative conclusion on the real results of the model evaluation.

7 DISCUSSION

7.1 Interpretation of findings

This system is an example of how machine learning can be employed in making decisions based on multiple variables.

However, for a good recommendation system, not only should it show good accuracy in prediction but also give recommendations that are comprehensible and applicable.

Thus, the effectiveness of the system will depend on not only the performance of the model but also on the quality of data provided.

It is important to interpret the results with caution since a machine learning prediction cannot guarantee success in cultivating.

7.2 Comparison with literature baselines and existing solutions

As compared to simple rule-based systems, the proposed system has scope for learning patterns from data related to crop cultivation.

As compared to environmental suitability-based approaches, the proposed approach tries to incorporate farm-specific conditions such as the size of the land and the budget of the farmer.

As compared to generic crop recommendation systems, the proposed component specifically looks at medicinal and herbal crops in the context of Sri Lanka.

The last comparison should be made using the actual figures found from the literature review.

7.3 Research contribution

The major contribution of this research is that of an AI/ML based cultivation recommendation system for small herbal farmers.

The research uses:

- Individual farmer information
- Environmental information
- Cultivation information

- Market considerations
- Economic viability
- Sustainability

This combination gives a wider perspective in decision support compared to recommendations based purely on one environment.

7.4 Product contribution

The Product contribution is an actual piece of software code which realizes the idea into the actual application.

The Flutter interface enables farmers to enter comprehensible data, whereas the Flask backend gives the functionality of machine-learning recommendation.

The design based on API integration also allows for integration of this component into other components of the system.

7.5 Limitations and threats to validity

A number of constraints must be taken into consideration.

Firstly, the accuracy of the prediction system relies on the nature and availability of the dataset.

Secondly, data about the cultivation of medicinal plants may not apply to all environmental conditions in Sri Lanka.

Thirdly, there might be changes in demand and value in the future.

Fourthly, predictions by machine learning methods do not guarantee successful cultivation.

Fifthly, weather data may rely on external sources and internet connection.

Sixthly, the current system does not take measurements of soil conditions using IoT sensors.

7.6 Implications

Implications of the system can be considered with regard to small farmers, agricultural decision-making, herb-based business, and future agricultural technology.

Based on AI and agricultural/economic data, the system might assist farmers in making better decisions regarding planting and cultivating crops.

However, suggestions provided by the system should be considered supportive information, and in an ideal world, supplemented by agricultural expertise.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The research created an AI-based cultivation recommendation system that is supposed to help small herbal farmers in Sri Lanka.

The issue that the system solves is the selection of suitable medicinal plants depending on various criteria such as soil type, size of land, budget for cultivation, environmental factors, demand for the product, economic feasibility and sustainability.

The software solution is based on the Flutter frontend and Flask backend.

The machine learning recommendation engine gives intelligence to the component and enables making cultivation decisions with the help of prediction algorithms.

The contribution of the component to the overall research project is in solving the problem of cultivation recommendation for intelligent management of Ayurvedic herbal plants.

8.2 Achievement of objectives

Objectives of the research have been achieved by means of identification of cultivation factors, creation of the cultivation dataset, training of machine learning algorithms, development of the Flutter interface, development of the Flask backend, model performance evaluation, and integration of the work into the overall research system.

The final submission will contain proof of all the objectives on the basis of experimental data, screenshots, and tests.

8.3 Individual research and product contributions

Research individual contribution will be on researching an AI-based system for recommending appropriate medicinal plants for small-scale farming communities.

Individual product contributions include:

- Recommendation Interface for cultivation

- Processing of inputs from farmers
- Budget validation
- Recommendation Flask API
- Machine Learning Prediction
- Evaluation of the model
- Integration into the overall application
- Software Testing and Documentation

8.4 Future directions

Further developments may involve increasing the data set by adding more information about medicinal plants and geographical locations in Sri Lanka.

The system can be further developed to add verified market price information, seasonal information, rainfall data, and climate data.

Further developments may include adding multilingual capabilities, especially those involving Sinhala and Tamil language interfaces.

Another area where further developments can occur involves the development of Explainable AI (Explainability), which will allow one to explain why a particular plant was suggested.

Another research area may involve exploring advanced machine learning algorithms like gradient boosting and ensemble techniques.

Testing the system with farmers in the field will provide further evidence on how effective the system actually is in practice.

In time, the system can be developed into a cloud-based agricultural decision support system, which will provide continuous updates based on verified environmental and market data.

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GLOSSARY

Artificial Intelligence (AI): A branch of computing which deals with creation of systems that are able to do things requiring human intelligence.

Cultivation Recommendation: A recommendation made by the computer system of suitable plants taking into account the cultivation constraints.

Feature: An input variable which is used by a machine learning algorithm to generate predictions.

Machine Learning: An approach to computing where algorithms learn patterns from data and generate predictions/decisions.

Medicinal Plant: A plant with properties/compounds which are used for medicinal purposes.

Small-Scale Farmer: A farmer who undertakes agricultural practices on a relatively small piece of land or with little financial means.

Random Forest: A machine learning algorithm which combines several decision trees to generate predictions.

Recommendation System: A software which analyzes the data and generates recommendations according to user needs.

APPENDICES

Appendix A Research instruments or extended results

A.1 Cultivation recommendation input fields

Input	Description
Soil type	Soil type available for farming
Land size	Size of land available for farming
Budget	The amount available for farming
Location/Weather	Environmental data related to the chosen location

Table 3: Cultivation recommendation input fields

Appendix B Academic integrity and AI use disclosure

Generative AI tools were employed as an auxiliary resource at selected points in the research and documentation process such as brainstorming, language enhancement, explanations of technical terms, and draft creation.

AI-aided materials must be reviewed and checked by the researcher prior to being included in the thesis. The responsibility of the researcher is to provide accurate, original information and experiments, references, data, results, analysis, and conclusions in the thesis.

Possible examples of AI-aided activities include:

- Enhancement of academic writing in terms of structure and clarity.
- Explanation of the programming terminology.
- Help with debugging and developing new ideas.
- Creation of draft documents.
- Propositions for machine learning implementation approaches.